3.7 Deep convolutional neural networks (AlexNet)

- now that we understand the basics of wiring together CNNs, we will take a tour of modern CNN architectures
- each next section will correspond to a significant CNN architecture that was briefly a dominant architecture and were winners (AlexNet, ResNet) or runners-up (VGG) in the ImageNet competition, which has served as a barometer of progress on supervised learning in computer vision since 2010
- these models include AlexNet, the first large-scale network deployed to beat conventional computer vision methods on a large-scale vision challenge; the VGG network, which makes use of a number of repeating blocks of elements; and residual networks (ResNet), which remain the most popular off-the-shelf architecture in computer vision

3.7 Deep convolutional neural networks (AlexNet)

- while the idea of *deep neural networks* is quite simple (stack together a bunch of layers), performance can vary wildly across architectures and hyperparameter choices
- the neural networks described next are the product of intuition, a few mathematical insights, and a whole lot of trial and error
- we present these models in chronological order, partly to convey a sense of the history, so that we can form our own intuitions about where the field is heading, and perhaps develop our own architectures
- for instance, batch normalization and residual connections, described next, have offered two popular ideas for training and designing deep models

3.7 Deep convolutional neural networks (AlexNet)

- although CNNs were well known in the computer vision and machine learning communities following the introduction of LeNet, they did not immediately dominate the field
- although LeNet achieved good results on early small datasets, the performance and feasibility of training CNNs on larger, more realistic datasets had yet to be established
- in fact, for much of the intervening time between the early 1990s and the breakthrough results of 2012, neural networks were often surpassed by other machine learning methods, such as support vector machines

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- for computer vision, this comparison is perhaps not fair
- that is, although the inputs to convolutional networks consist of raw or lightly-processed (e.g., by centering) pixel values, practitioners would never feed raw pixels into traditional models
- instead, typical computer vision pipelines consisted of *manually* engineering *feature extraction* pipelines
- rather than *learn the features*, the features were *crafted*
- most of the progress came from having more clever ideas for features, and the learning algorithm was often very simple

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- although some neural network accelerators were available in the 1990s, they were not yet sufficiently powerful to make deep multichannel, multilayer CNNs with a large number of parameters
- moreover, datasets were still relatively small
- added to these obstacles, key tricks for training neural networks including parameter initialization heuristics, clever variants of stochastic gradient descent, non-squashing activation functions, and effective regularization techniques were still missing
- thus, rather than training *end-to-end* (pixel to classification) systems, classical pipelines looked more like this:
  1. Obtain an interesting dataset. In early days, these datasets required expensive sensors (at the time, 1 megapixel images were state-of-the-art).
  2. Preprocess the dataset with hand-crafted features based on some knowledge of optics, geometry, other analytic tools, and occasionally on the discoveries of lucky graduate students.
  3. Feed the data through a standard set of feature extractors such as the SIFT (scale-invariant feature transform), the SURF (speeded up robust features), or any number of other hand-tuned pipelines.
  4. Dump the resulting representations into a classifier, likely a linear model or kernel method, to train a classifier.

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- another group of researchers, including Yann LeCun, Geoffrey Hinton, Yoshua Bengio, Andrew Ng, and Juergen Schmidhuber had different plans: they believed that features themselves should be learned
- moreover, they believed that, to be reasonably complex, the features should be *hierarchically* composed with multiple jointly learned layers, each with learnable parameters
- in the case of an image, the lowest layers might come to detect edges, colors, and textures
- indeed, Alex Krizhevsky, Ilya Sutskever, and Geoffrey Hinton proposed a new variant of a CNN, *AlexNet*, that achieved excellent performance in the 2012 ImageNet challenge
- AlexNet was named after Alex Krizhevsky, the first author of the breakthrough ImageNet classification paper

3.7 Deep convolutional neural networks (AlexNet)

- interestingly, in the lowest layers of the network, the model learned feature extractors that resembled some traditional filters; Figure 11 is reproduced from the AlexNet paper, and describes lower-level image descriptors

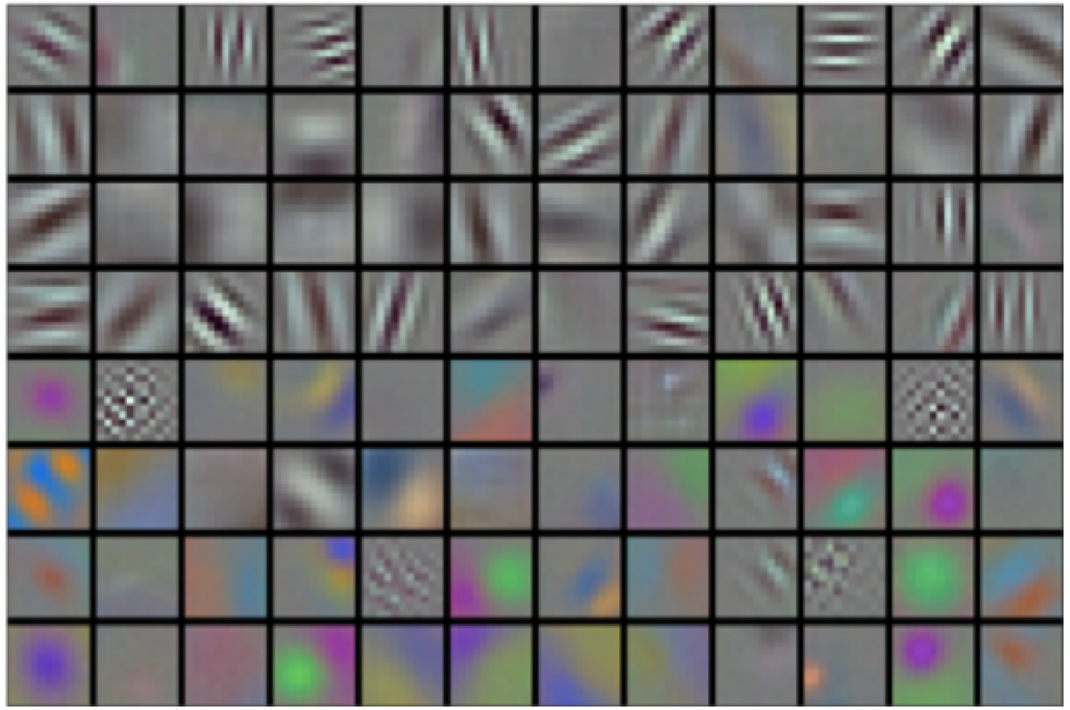

Figure 11: Image filters learned by the first layer of AlexNet.

3.7 Deep convolutional neural networks (AlexNet)

- higher layers in the network might build upon these representations to represent larger structures, like eyes, noses, blades of grass, and so on
- even higher layers might represent whole objects like people, airplanes, dogs, or frisbees
- ultimately, the final hidden state learns a compact representation of the image that summarizes its contents, such that data belonging to different categories can be easily separated
- while the ultimate breakthrough for many-layered CNNs came in 2012, a core group of researchers had dedicated themselves to this idea, attempting to learn *hierarchical representations* of visual data for many years
- the ultimate breakthrough in 2012 can be attributed to *two key factors*

3.7 Deep convolutional neural networks (AlexNet)

- deep models with many layers require large amounts of *data* in order to enter the regime where they significantly outperform traditional methods based on convex optimizations (e.g., linear and kernel methods)
- however, given the limited storage capacity of computers, the relative expense of sensors, and the comparatively tighter research budgets in the 1990s, most research relied on tiny datasets
- numerous papers addressed the UCI collection of datasets, many of which contained only hundreds or (a few) thousands of images captured in unnatural settings with low resolution

3.7 Deep convolutional neural networks (AlexNet)

- in 2009, the ImageNet dataset was released, challenging researchers to learn models from 1 million examples, each from one of 1000 distinct categories of objects
- the researchers, led by Fei-Fei Li, who introduced this dataset, leveraged Google Image Search to prefilter large candidate sets for each category and employed the Amazon Mechanical Turk crowdsourcing pipeline to confirm for each image whether it belonged to the associated category
- this scale was unprecedented
- the associated competition, named the ImageNet Challenge pushed computer vision and machine learning research forward, challenging researchers to identify which models performed best at a greater scale than academics had previously considered

3.7 Deep convolutional neural networks (AlexNet)

- the second factor is *hardware*
- deep learning models are massive consumers of compute cycles
- training can take hundreds of epochs, and each iteration requires passing data through many layers of computationally-expensive linear algebra operations
- this is one of the main reasons why, in the 1990s and early 2000s, simple algorithms based on the more-efficiently optimized convex objectives were preferred

3.7 Deep convolutional neural networks (AlexNet)

- *graphical processing units* (GPUs) proved to be a game changer in making deep learning feasible
- these chips had long been developed for accelerating graphics processing to benefit computer games
- in particular, they were optimized for high throughput 4×4 matrix-vector products, which are needed for many computer graphics tasks
- fortunately, this math is strikingly similar to that required to calculate convolutional layers
- around that time, NVIDIA and ATI had begun optimizing GPUs for general computing operations, going as far as to market them as *general-purpose GPUs* (GPGPU)

3.7 Deep convolutional neural networks (AlexNet)

- back to 2012; a major breakthrough came when Alex Krizhevsky and Ilya Sutskever implemented a deep CNN that could run on GPU hardware
- they realized that the computational bottlenecks in CNNs, convolutions and matrix multiplications, are all operations that could be parallelized in hardware
- using two NVIDIA GTX 580s with 3GB of memory, they implemented fast convolutions
- the code `cuda-convnet` was good enough that, for several years, it was the industry standard, and powered the first couple years of the deep learning boom

3.7 Deep convolutional neural networks (AlexNet)

- AlexNet, which employed an 8-layer CNN, won the ImageNet Large Scale Visual Recognition Challenge 2012 by a phenomenally large margin
- this network showed, for the first time, that the features obtained by learning can transcend manually-designed features, breaking the previous paradigm in computer vision
- the architectures of AlexNet and LeNet are very similar, as Figure 12 illustrates
- note that we provide a slightly streamlined version of AlexNet, removing some of the design tricks that were needed in 2012 to make the model fit on two small GPUs

3.7 Deep convolutional neural networks (AlexNet)

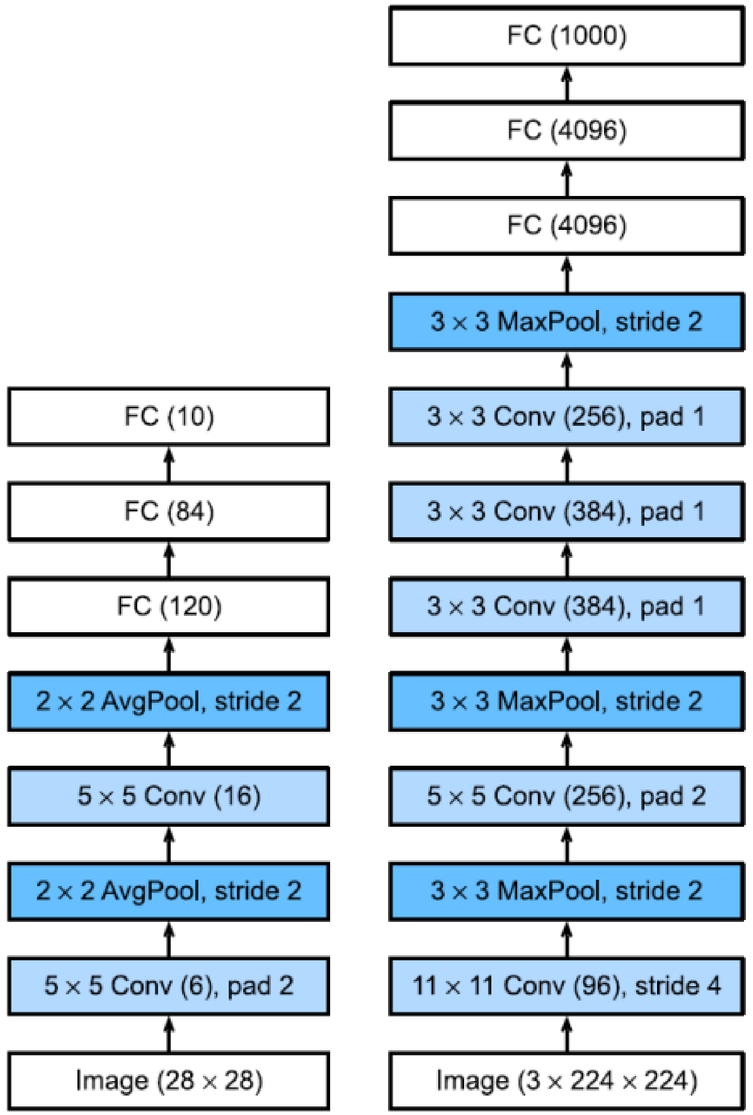

Figure 12: From LeNet (left) to AlexNet (right).

3.7 Deep convolutional neural networks (AlexNet)

- the design philosophies of AlexNet and LeNet are very similar, but there are also significant differences
- first, AlexNet is much deeper than the comparatively small LeNet5
- AlexNet consists of eight layers: five convolutional layers, two fully-connected hidden layers, and one fully-connected output layer
- second, AlexNet used the ReLU instead of the sigmoid as its activation function

3.7 Deep convolutional neural networks (AlexNet)

- in AlexNet's first layer, the convolution window shape is 11×11
- since most images in ImageNet are more than ten times higher and wider than the MNIST images, objects in ImageNet data tend to occupy more pixels
- consequently, a larger convolution window is needed to capture the object
- the convolution window shape in the second layer is reduced to 5×5, followed by 3×3
- in addition, after the first, second, and fifth convolutional layers, the network adds maximum pooling layers with a window shape of 3×3 and a stride of 2
- moreover, AlexNet has ten times more convolution channels than LeNet

3.7 Deep convolutional neural networks (AlexNet)

- after the last convolutional layer there are two fully-connected layers with 4096 outputs
- these two huge fully-connected layers produce model parameters of nearly 1 GB
- due to the limited memory in early GPUs, the original AlexNet used a dual data stream design, so that each of their two GPUs could be responsible for storing and computing only its half of the model
- fortunately, GPU memory is comparatively abundant now, so we rarely need to break up models across GPUs these days (our version of the AlexNet model deviates from the original paper in this aspect)

3.7 Deep convolutional neural networks (AlexNet)

- besides, AlexNet changed the sigmoid activation function to a simpler ReLU activation function
- on one hand, the computation of the ReLU activation function is simpler; for example, it does not have the exponentiation operation found in the sigmoid activation function
- on the other hand, the ReLU activation function makes model training easier when using different parameter initialization methods
- this is because, when the output of the sigmoid activation function is very close to 0 or 1, the gradient of these regions is almost 0, so that backpropagation cannot continue to update some of the model parameters
- in contrast, the gradient of the ReLU activation function in the positive interval is always 1
- therefore, if the model parameters are not properly initialized, the sigmoid function may obtain a gradient of almost 0 in the positive interval, so that the model cannot be effectively trained

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- AlexNet controls the model complexity of the fully-connected layer by dropout (Section 2.5), while LeNet only uses weight decay
- to augment the data even further, the training loop of AlexNet added a great deal of image augmentation, such as flipping, clipping, and color changes
- this makes the model more robust, and the larger sample size effectively reduces overfitting